Question 1 (Basic Commands)

Which command is used in Linux to display the current working directory?

Answer Choices: A. ls B. cd C. pwd D. man

Question 2 (Linux vs. Unix)

Linux is an operating system that is a _____ of UNIX.

Answer Choices: A. Direct commercial fork B. Proprietary successor C. Freely distributed clone/re-implementation D. Complete rewrite with no shared code

Question 3 (File System Permissions)

What is the numeric (octal) permission that represents read and write access for a user, but no access for group and others?

Answer Choices: A. 700 B. 644 C. 600 D. 777

Question 4 (Shell Scripting Basics)

What is the typical first line (shebang) of a Bash shell script?

Answer Choices: A. //bin/bash B. #!/bin/sh C. #!bash D. #!/bin/bash

Question 5 (Package Managers)

Which package manager is primarily used on Debian-based Linux distributions like Ubuntu?

Answer Choices: A. RPM B. DNF C. YUM D. APT

Question 6 (Process Basics)

A running instance of a program is referred to as a:

Answer Choices: A. Kernel B. Daemon C. Process D. Shell

Correct Answer: C. Process

Question 7 (Memory Management)

The technique of dividing a program's logical address space into fixed-size blocks and loading them into non-contiguous physical memory is known as:

Answer Choices: A. Segmentation B. Context Switching C. Virtualization D. Paging

Correct Answer: D. Paging

Question 8 (Scheduling)

Which CPU scheduling algorithm is non-preemptive and selects the process with the smallest execution time next?

Answer Choices: A. Round Robin (RR) B. First-Come, First-Served (FCFS) C. Shortest Job First (SJF) D. Priority Scheduling

Question 9 (Shell Scripting - Loops)

Which of the following loops will execute *at least once*, regardless of the initial condition, in a traditional programming sense (though not explicitly a Bash construct)?

Answer Choices: A. for loop B. while loop C. until loop D. None of the above (Bash loops check condition before first iteration)

Question 9 (Shell Scripting - Loops)

Which of the following loops will execute *at least once*, regardless of the initial condition, in a traditional programming sense (though not explicitly a Bash construct)?

Answer Choices: A. for loop B. while loop C. until loop D. None of the above (Bash loops check condition before first iteration)

Question 10 (File Systems)

Which Linux file system is an improvement over Ext3, offering better performance, support for larger file sizes, and delayed allocation?

Answer Choices: A. XFS B. Ext2 C. FAT32 D. Ext4

Question 11 (Shell Scripting - Conditional)

In a Bash if statement, which of the following is the correct syntax for checking if a variable $count is greater than or equal to 10?

Answer Choices: A. if [$count >= 10] B. if [[$count -ge 10]]; then C. if ($count > 9) D. if $count -gte 10

Question 12

Question 13 (Process States)

When a process is waiting for an I/O operation (like disk read) to complete, it is typically in which process state?

Answer Choices: A. New B. Running C. Ready D. Waiting/Blocked

Question 14 (Memory Management - Virtual Memory)

What is the primary purpose of **Virtual Memory** in an operating system?

Answer Choices: A. To provide faster access to physical RAM using a cache. B. To allow a process to execute even if the entire process is not in physical memory. C. To prevent memory fragmentation by using fixed-size pages. D. To implement inter-process communication (IPC) through shared memory segments.

Question 15 (Synchronization)

A **critical section** in a program is a code segment that:

Answer Choices: A. Must be executed by a process at the highest priority. B. Accesses a shared resource and must be executed atomically (by only one process at a time). C. Contains only conditional statements and loops. D. Allocates memory for the process.

Question 16 (Package Managers)

What is the command to list all currently installed RPM packages on a system?

Answer Choices: A. yum list all B. dpkg -l C. rpm -qa D. apt list --installed

Question 17 (File System Permissions - chmod)

What is the effect of the command chmod o+w my_file.txt?

Answer Choices: A. Remove write permission from the owner of my_file.txt. B. Grant write permission to the group of my_file.txt. C. Grant write permission to others for my_file.txt. D. Set the execute permission for all users of my_file.txt.

Question 18 (Shell Scripting - Loops)

Which Bash loop structure is used to iterate over a list of items (e.g., file names, strings, numbers)?

Answer Choices: A. while loop B. until loop C. for loop D. if loop

Correct Answer: C. for loop

Question 19 (Scheduling Algorithms)

A major problem with the non-preemptive Shortest Job First (SJF) scheduling algorithm is:

Answer Choices: A. Excessive context switching overhead. B. Determining the length of the next CPU burst. C. Short jobs waiting indefinitely (starvation). D. Its complexity compared to FCFS.

Question 20 (Process Control Block)

What critical information is *not* typically stored in a Process Control Block (PCB)?

Answer Choices: A. Process state B. Program Counter C. CPU Scheduling information D. The complete executable code of the program

Question 21 (Basic Commands - Redirection)

What is the purpose of the 2> operator in a Bash command?

Answer Choices: A. Redirect standard input from a file. B. Append standard output to a file. C. Redirect standard error (stderr) to a file. D. Pipe the output of one command to the input of another.

Question 22 (Synchronization Techniques)

Which synchronization tool is an integer variable that is accessed only through two atomic operations, wait() and signal()?

Answer Choices: A. Mutex B. Race condition C. Semaphore D. Thread

Question 23 (Virtual Memory)

The process of bringing a page of a program from secondary storage (disk) into physical memory is called:

Answer Choices: A. Segmentation fault B. Context switch C. Thrashing D. Page fault

Correct Answer: D. Page fault

Question 24 (Linux File System Types)

What does the acronym **XFS** stand for, which is a high-performance journaling file system often used for large file servers?

Answer Choices: A. eXecutable File System B. Extended File Standard C. X-Server File Structure D. Extents File System

Question 25 (Shell Scripting - until loop)

An until loop in Bash continues to execute as long as the conditional test provided is:

Answer Choices: A. True B. False C. Greater than zero D. Zero (success)

Question 26 (Linux vs. Unix)

A core difference in development philosophy between the Linux kernel and traditional UNIX kernels is that Linux is developed:

Answer Choices: A. Exclusively by a single company (vendor-specific). B. Under a closed-source, proprietary model. C. Collaboratively and publicly under an open-source model. D. Primarily for mainframe computers.

Question 27 (Package Managers)

Which command is used on a Red Hat-based system (like Fedora/CentOS 8-Stream) to install a package, replacing the older yum command as the default?

Answer Choices: A. apt install B. rpm -i C. dnf install D. zypper install

Question 28 (Process Types)

What is a **daemon** process?

Answer Choices: A. A process that requires constant user input to run. B. A foreground process that runs in the current terminal session. C. A background service process that runs without a controlling terminal. D. A process that has been suspended by the operating system.

Question 29 (Memory Allocation)

Which memory allocation technique can suffer from **external fragmentation**, where there is enough total free space for a process, but it is not contiguous?

Answer Choices: A. Paging B. Segmentation C. Virtual Memory D. Swapping

Question 30 (Scheduling Algorithms)

In a **Round Robin (RR)** CPU scheduling algorithm with a time quantum of Q, what happens when a process finishes its CPU burst before the time quantum expires?

Answer Choices:

A. The remaining time in the quantum is simply wasted.

B. The process is immediately moved to the ready queue.

C. The CPU is immediately preempted and assigned to the next process.

D. The process is paused until the next quantum.

Question 31 (Synchronization - Deadlock)

Which one of the following is **NOT** a necessary condition for a **Deadlock** to occur in a system?

Answer Choices: A. Mutual Exclusion B. Circular Wait C. Preemption D. Hold and Wait

Question 32 (Context Switching)

Which of the following operations contributes the most to the overhead (time cost) of a **Context Switch**?

Answer Choices: A. Updating the process state in the PCB. B. Loading the new Program Counter (PC). C. Invalidating and flushing the Translation Lookaside Buffer (TLB). D. Moving the process from the Running state to the Ready state.

Question 33 (Shell Scripting - Command Line Arguments)

Consider the Bash script line: shift 2. If the script was executed as ./script.sh arg1 arg2 arg3 arg4, what will be the value of $1 after this line executes?

Answer Choices: A. arg1 B. arg2 C. arg3 D. arg4

Question 34 (Memory Management - Paging)

In a paging system, if the logical address is 32 bits, the page size is 4KB (2^{12} bytes), how many entries are there in a single-level page table for a process?

Answer Choices:

A. 2^{12}

B. 2^{16}

C. 2^{20}

D. 2^{32}

Question 35 (File System - Mounting)

In Linux, what is the significance of the **mount point**?

Answer Choices: A. It is the location on the disk where the filesystem's superblock is stored. B. It is the directory in the existing file system where the root of the new file system will be attached. C. It is the command used to check the integrity of a file system. D. It is the device name of the partition being formatted.

Question 36 (Synchronization - Mutex vs. Semaphore)

What is the primary difference between a **Mutex Lock** and a **Counting Semaphore**?

Answer Choices: A. A mutex is used for signaling between processes, while a semaphore is for mutual exclusion. B. A mutex can be acquired by multiple threads simultaneously, while a semaphore only allows one. C. A mutex is typically binary (0 or 1) and is used for mutual exclusion, while a counting semaphore can have a value greater than 1 and is used for resource counting. D. A semaphore is always implemented in user space, and a mutex is always implemented in the kernel.

Question 37 (CPU Scheduling)

Which of the following scheduling algorithms can lead to the **Convoy Effect**, where a short process is forced to wait for a single large process to finish?

Answer Choices: A. Shortest Remaining Time First (SRTF) B. First-Come, First-Served (FCFS) C. Priority Scheduling with preemption D. Round Robin (RR)

Question 38 (Shell Scripting - if-elif-else)

In a shell script, which test is used inside if [...] to check if a file named config.txt exists **AND** is readable?

Answer Choices: A. [-e config.txt -a -r config.txt] B. [-a config.txt -r config.txt] C. [-f config.txt] && [-x config.txt] D. [-r config.txt && -f config.txt]

Question 39 (Package Managers - Manual Install)

When manually installing a software package using a package manager file (e.g., an .rpm file), why is the manual installation generally discouraged compared to using the manager's front-end tool (like dnf or apt)?

Answer Choices: A. Manual install uses up more disk space. B. Manual install automatically removes conflicting dependencies. C. Manual install bypasses dependency checking and resolution. D. Manual install can only be done as the root user.

Question 40 (Linux vs. UNIX - Kernel Architecture)

The Linux kernel uses a **monolithic architecture** model. Which of the following is a characteristic of a monolithic kernel?

Answer Choices: A. Core services (like device drivers, file system) run in user space. B. All core OS services are compiled into a single, large kernel image and run in kernel space. C. The kernel's functionality can be easily removed or added during runtime. D. The kernel is highly modular and fails gracefully if a component crashes.